Classical Field Theory — Exercise Sheet 24

A higher Lax flow and operator domains

Questions only

Use $D_x = \partial_x$ and the ordered commutator $[B, L] = BL - LB$. All operator calculations in the first problem are on $C_c^\infty(\mathbb{R}, \mathbb{C})$, with smooth real coefficient u. A displayed differential expression does not specify its Hilbert-space domain.

Exercise 24.1 (Determining a fifth-order Lax flow). Let $L = -D_x^2 + u$. For real constants $r_1, \dots, r_6$ consider

$$B_5 = 16D_x^5 + r_1 u D_x^3 + r_2 u' D_x^2 + (r_3 u'' + r_4 u^2) D_x + r_5 u''' + r_6 u u'.$$

Products with u and its derivatives here mean multiplication before the indicated derivative of the test.

(a) Determine the constants so that $[B_5, L]$ is multiplication for every smooth u. Find the resulting scalar coefficient and write the corresponding evolution equation $L_t = [B_5, L]$.

(b) Prove that the resulting B_5 is formally skew-symmetric on compact smooth tests. Distinguish this from a statement about a closed skew-adjoint operator on L^2.

Exercise 24.2 (Constancy for a sufficiently regular eigenbranch). Let $u \in C^1(I; \mathcal{S}(\mathbb{R}, \mathbb{R}))$ solve $u_t = K_5[u]$ on an open interval I. Suppose $\lambda \in C^1(I; \mathbb{R})$ and $g \in C^1(I; H^7(\mathbb{R}, \mathbb{C}))$, with $g(t) \neq 0$ and $L(t)g(t) = \lambda(t)g(t)$ at every time. The Sobolev space H^7 uses the Fourier norm $\|\langle k \rangle^7 \widehat{g}\|_2$.

Use the following operator fact as a *given*: a real bounded multiplication perturbation of $-D_x^2$ is self-adjoint on the common domain $H^2(\mathbb{R})$. This is the relative-bound-zero case of Kato, *Perturbation Theory for Linear Operators*, V, Theorem 4.3, p. 287. Prove that λ is constant. Specify the domain needed for moving L across an inner product. Does this calculation require simplicity or isolation, and does it construct a branch or prove a theorem about the whole spectrum?

Exercise 24.3 (The same interior expression with moving endpoints). Fix $\ell > 0$ and a smooth positive nonconstant function $\kappa(t)$. On $L^2(0, \ell)$ consider the expression $L = -\partial_x^2$ with domain

$$\mathcal{D}_t = \left\{ f \in H^2(0, \ell) : f(0) = 0, \quad f'(\ell) = \beta(t) f(\ell) \right\}, \qquad \beta(t) = \kappa(t) \coth(\kappa(t)\ell).$$

The trace values in this formula have their usual continuous Sobolev representatives on a finite interval.

(a) Show that this realization is symmetric by its endpoint Green formula. Find a nonzero smooth eigenfunction with a nonconstant eigenvalue. Self-adjointness is not required for this task.

(b) With $B = 0$, verify $L_t \psi = [B, L]\psi = 0$ on every compactly supported interior test. Explain why this does not contradict the preceding conditional eigenbranch result or establish constancy of the varying eigenvalue.